

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks: 50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total (10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Kongara Jogesh (192525035) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: k.jogesh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

	User Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

	Farm Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

	Employee Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

CSA16 – DATA WAREHOUSING AND DATA MINING
ASSESSMENT TOOL 1 – ANSWERS

Q1. Hospital Patient Segmentation – K-Means Clustering

Understanding of Data

The dataset contains five patients described by monthly medical expense and health risk score. The provided SSE values decrease from 600 at K=1 to 190 at K=5. The largest improvements occur from K=1 to K=2 and from K=2 to K=3, after which the reduction becomes small.

K	SSE	Reduction from previous K
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

Analysis

The elbow point is K=3. Up to three clusters, increasing K produces a substantial improvement in compactness, shown by the sharp SSE reduction. From K=3 onward, the improvement is only 20 and then 10, so adding more clusters provides relatively little additional benefit. Therefore, K=3 gives a good balance between model simplicity and within-cluster similarity.

Application of Concepts

With three clusters, the patients can be interpreted as three broad healthcare segments. P1 and P4 have relatively low medical expenses and high risk scores, so they can represent a high-risk, lower-cost segment. P3 has a higher risk score together with a mid-range expense, making it a distinct high-risk/intermediate-expense segment. P2 and P5 have higher expenses but lower risk scores, forming a

higher-cost, lower-risk segment. These groupings are an interpretation of the supplied values; the assessment does not provide final K-Means centroids.

Conclusion

The optimal number of clusters is 3. Selecting $K=3$ allows the healthcare organization to separate patients into meaningful groups without creating unnecessary clusters. This can support differentiated monitoring, resource allocation, preventive care planning, and cost-management decisions. The three-cluster solution is preferable because the SSE improvement after $K=3$ is minimal.

Clarity

The SSE table and chart clearly show the elbow at $K=3$, directly supporting the decision.

Q2. E-Commerce Order Analysis – Hierarchical Clustering

Understanding of Data

The dataset contains five orders with product category and order value. Electronics orders are O1 (₹55,000) and O3 (₹57,000). Clothing orders are O2 (₹25,000) and O5 (₹27,000). Furniture is represented by O4 (₹30,000). The question states that a dendrogram is generated, but the actual dendrogram is not included in the supplied assessment pages.

Order ID	Category	Order Value (₹)
O1	Electronics	55,000
O2	Clothing	25,000
O3	Electronics	57,000
O4	Furniture	30,000
O5	Clothing	27,000

Analysis

Hierarchical clustering progressively joins observations according to similarity and represents those joins in a dendrogram. From the supplied values, O1 and O3 are highly similar because both belong to Electronics and have close order values. O2 and O5 are also highly similar because both belong to Clothing and have close order values. O4 is a separate Furniture order with a value close to the Clothing orders but a different category.

The most interpretable grouping from the supplied dataset is therefore three clusters: an Electronics cluster (O1, O3), a Clothing cluster (O2, O5), and a Furniture cluster (O4). However, the exact number of clusters selected by cutting a hierarchical dendrogram cannot be verified from the source because the dendrogram and its linkage-distance scale are not provided. Thus, three clusters is the data-based interpretation rather than a claim about an unseen dendrogram.

Application of Concepts

The three-cluster interpretation can help the retailer understand purchasing behavior by separating high-value electronics purchasing from lower-value clothing purchasing and the distinct furniture order. Electronics shows both high order value and internal consistency. Clothing shows lower and closely grouped values. Furniture needs separate observation because only one order is available in the dataset.

Conclusion

A three-cluster interpretation is the most meaningful based on the provided records, but the assessment's actual dendrogram is required to confirm the precise cut level. These clusters can help with category-specific marketing, inventory planning, customer segmentation, and pricing analysis.

Clarity

The order table makes the similarity within Electronics and Clothing immediately visible and explains why those pairs would be joined early in hierarchical clustering.

Q3. Mobile App User Behavior – Probability-Based / Soft Cluster Membership

Understanding of Data

The dataset describes five users using three variables: daily usage time, login frequency, and in-app purchase amount. U1, U3, and U5 have high usage times and high login frequencies, while U2 and U4 have lower usage and login frequencies but higher purchase amounts.

User	Daily Usage (min)	Login Frequency	Purchase (₹)
U1	120	18	2,000
U2	60	5	5,000
U3	140	22	1,500
U4	50	4	6,000
U5	130	20	1,800

Analysis

The question asks for probability-based membership values, but numerical membership probabilities are not supplied in the assessment. Therefore, exact probabilities cannot be calculated or assigned from the source alone. Conceptually, a soft-clustering method gives each user a degree of membership in multiple clusters rather than forcing a single absolute assignment.

From the observed behavior, U1, U3, and U5 strongly resemble a high-engagement segment because their usage times and login frequencies are high. U2 and U4 resemble a lower-engagement but higher-purchase segment. A borderline user could receive meaningful membership in both segments if their behavior lies between them; this is the key interpretation of probability-based membership.

Application of Concepts

Soft clustering is useful because customer behavior is not always cleanly separated. For example, a user may have moderate usage but unusually high purchase activity. Instead of placing that user completely into one segment, soft clustering can represent partial membership in more than one behavioral group. This can improve personalization, campaign targeting, retention analysis, and customer-value interpretation.

Advantages of Soft Clustering over Hard Clustering

- Represents uncertainty instead of forcing every user into exactly one group.
- Captures overlapping customer behaviors and transition patterns.
- Supports more flexible marketing and recommendation decisions.
- Provides richer information when user behavior does not form sharply separated groups.

Conclusion

Soft clustering is advantageous for mobile-app behavior because users can exhibit characteristics of multiple segments. The supplied dataset supports identifying high-engagement and high-purchase patterns, but exact probability membership values require the model output, which is not included in the assessment.

Clarity

The interpretation distinguishes observed user patterns from unavailable numerical membership probabilities, avoiding unsupported values.

Q4. Agricultural Crop Monitoring – Effect of Normalization

Understanding of Data

The farm dataset contains rainfall in millimetres, soil fertility score, and crop yield in tons. Their numerical ranges are very different: rainfall ranges from 280 to 520, fertility from 3 to 9, and yield from 65 to 90.

Farm	Rainfall (mm)	Soil Fertility	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Analysis

Before normalization, a distance-based clustering algorithm can be strongly influenced by rainfall because its numerical scale is much larger than soil fertility. Differences of hundreds of millimetres can dominate differences in the fertility score. This can cause clusters to reflect rainfall more than the other agricultural characteristics.

After normalization, the variables are transformed to comparable scales. Consequently, rainfall, soil fertility, and crop yield can contribute more evenly to distance calculations. This makes the resulting clusters more representative of combined agricultural conditions rather than being dominated by the variable with the largest numerical range.

Application of Concepts

For example, F4 has low rainfall, high fertility, and high yield, while F5 has high rainfall, low fertility, and low yield. Without normalization, the rainfall difference between these farms can have a large effect on distance. After normalization, the algorithm can also give appropriate importance to their fertility and yield differences. Similarly, F1, F2, and F3 can be evaluated using all three attributes rather than primarily rainfall.

Comparison

Aspect	Before Normalization	After Normalization
Feature scale	Very different	Comparable
Dominant variable risk	High; rainfall can dominate	Reduced
Cluster interpretation	May emphasize rainfall	More balanced across variables
Reliability	Sensitive to measurement scale	More suitable for distance-based clustering

Conclusion

Normalization improves the reliability of agricultural clustering because it reduces the effect of unequal measurement scales. It helps the algorithm identify clusters using rainfall, fertility, and crop yield together, producing a more balanced basis for agricultural monitoring and decision-making.

Clarity

The before-versus-after comparison directly connects the numerical scale problem to its effect on cluster formation.

Q5. Employee Performance Evaluation – Silhouette Analysis

Understanding of Data

The employee dataset contains productivity score, attendance percentage, and skill assessment score. Three clustering models are evaluated using silhouette scores: K=2 gives 0.48, K=3 gives 0.68, and K=4 gives 0.52.

Model	Silhouette Score	Interpretation
K=2	0.48	Moderate separation
K=3	0.68	Best separation among supplied models
K=4	0.52	Lower than K=3

Analysis

The silhouette score measures how well observations fit within their own cluster compared with other clusters. Among the supplied models, K=3 has the highest score, 0.68. It is higher than K=2 at 0.48 and K=4 at 0.52. Therefore, K=3 provides the strongest clustering structure among the tested choices.

Application of Concepts

Using three clusters can help the company distinguish employee performance profiles across productivity, attendance, and skill. The exact employee-to-cluster assignments are not provided by the assessment, so they should not be invented. After selecting K=3, the company can examine the cluster centroids or cluster profiles to label groups such as stronger overall performance, moderate performance, and areas requiring development, provided those profiles are supported by the model output.

A higher silhouette score at K=3 also indicates better separation and cohesion than the alternatives tested. This makes the three-cluster model more suitable for subsequent performance evaluation and targeted organizational actions.

Conclusion

The most suitable number of clusters is $K=3$ because it has the highest supplied silhouette score of 0.68. This model offers the best balance of within-cluster similarity and between-cluster separation among $K=2$, $K=3$, and $K=4$. It can support more meaningful employee performance evaluation when combined with cluster profiles and appropriate organizational interpretation.

Clarity

The silhouette table and chart make the selection of $K=3$ transparent and directly support the conclusion.

Overall Rubric Alignment

Understanding of Data (25%): Each answer identifies the variables, patterns, ranges, or supplied model values before drawing conclusions.

Analysis (25%): Each answer explains why a particular cluster choice or preprocessing interpretation is appropriate, including numerical comparisons where the source provides them.

Application of Concepts (20%): K-Means, hierarchical clustering, soft/probability-based membership, normalization, and silhouette analysis are connected to the given real-world datasets.

Conclusion (20%): Each question ends with a clear decision or implication. Where the source lacks a required output, the limitation is explicitly stated rather than replaced with an unsupported value.

Clarity (10%): The assessment is organized question-by-question with headings, tables, and black-only charts. The formatting uses Times New Roman, 12-point text throughout the answer section.

Note on source completeness

The supplied assessment contains the datasets, SSE values, and silhouette scores, but it does not include the actual dendrogram for Q2 or numerical probability-membership values for Q3. The answers therefore interpret those questions from the information actually supplied and explicitly identify where exact model outputs are unavailable.